Hellenic Complex Systems Laboratory

Exact Confidence Intervals for a Single Proportion

Technical Report XVI

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Short Description of the Demonstration

This Demonstration provides point estimates and exact confidence intervals for single population proportions defined by the presence or absence of a specified condition or trait. It also plots the corresponding lower and upper confidence limits as functions of the selected confidence level.

Search Terms

proportion, confidence interval, exact method, *F* distribution, inference

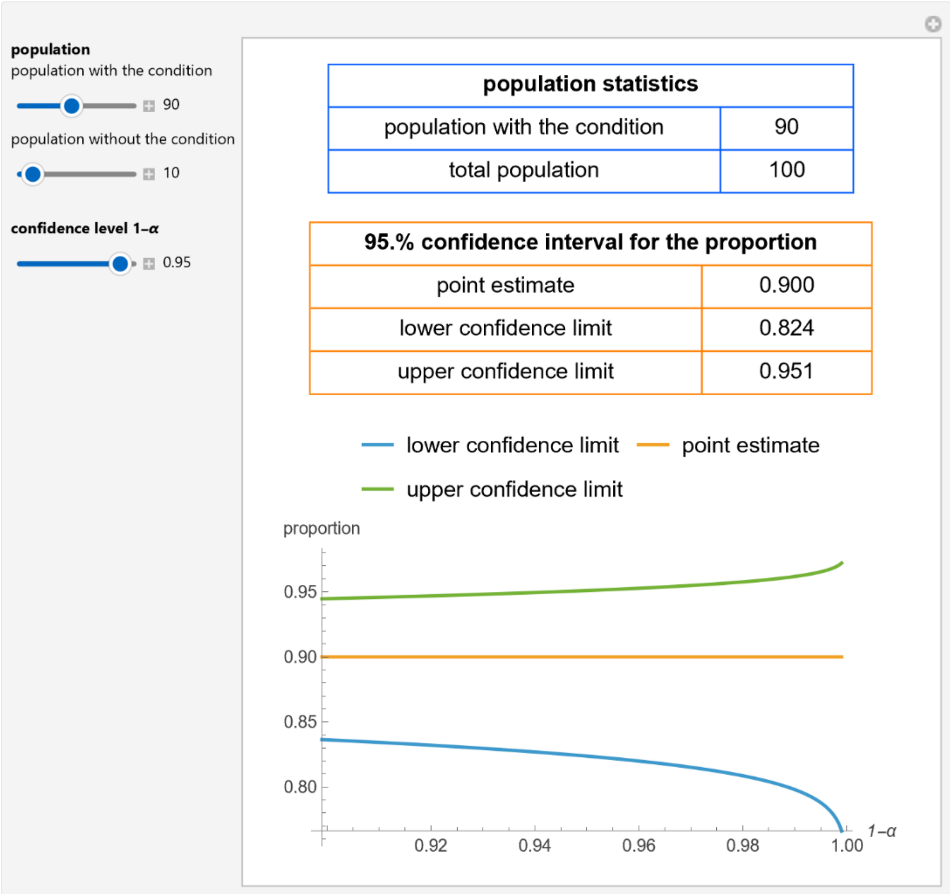


Figure 1: *Population statistics, point estimate, and 95% confidence interval for a single proportion of a population obeying a condition, as well as plots of the confidence limits versus confidence level, with the settings shown at the left.*

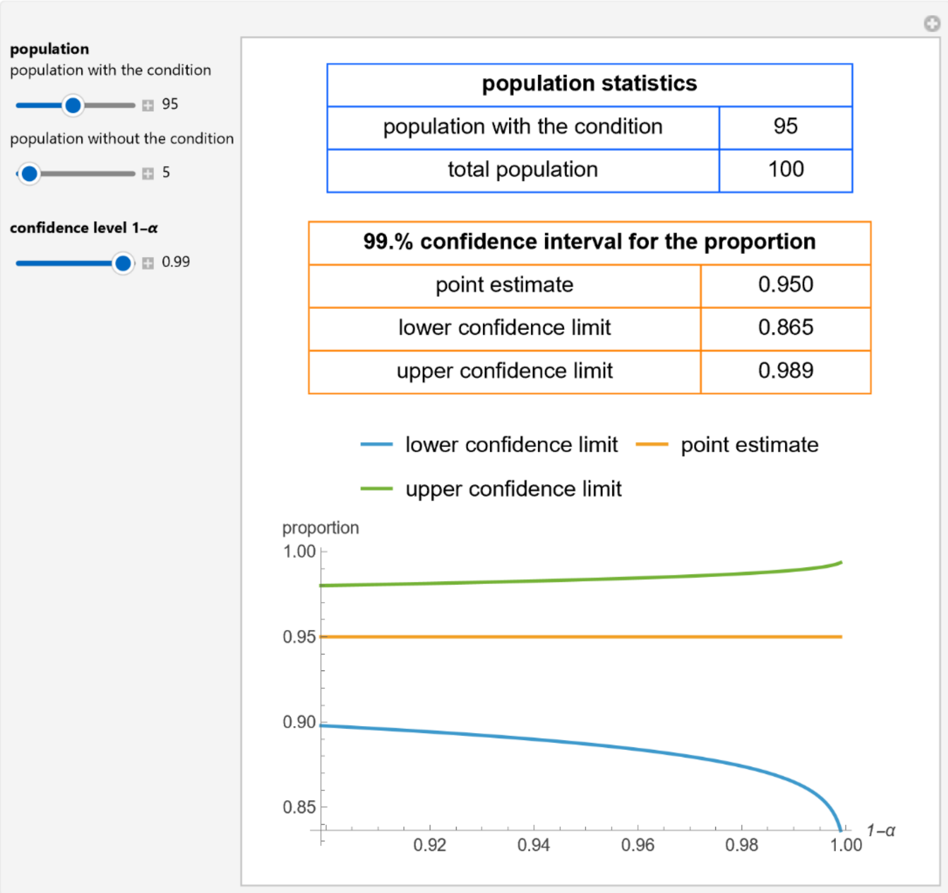
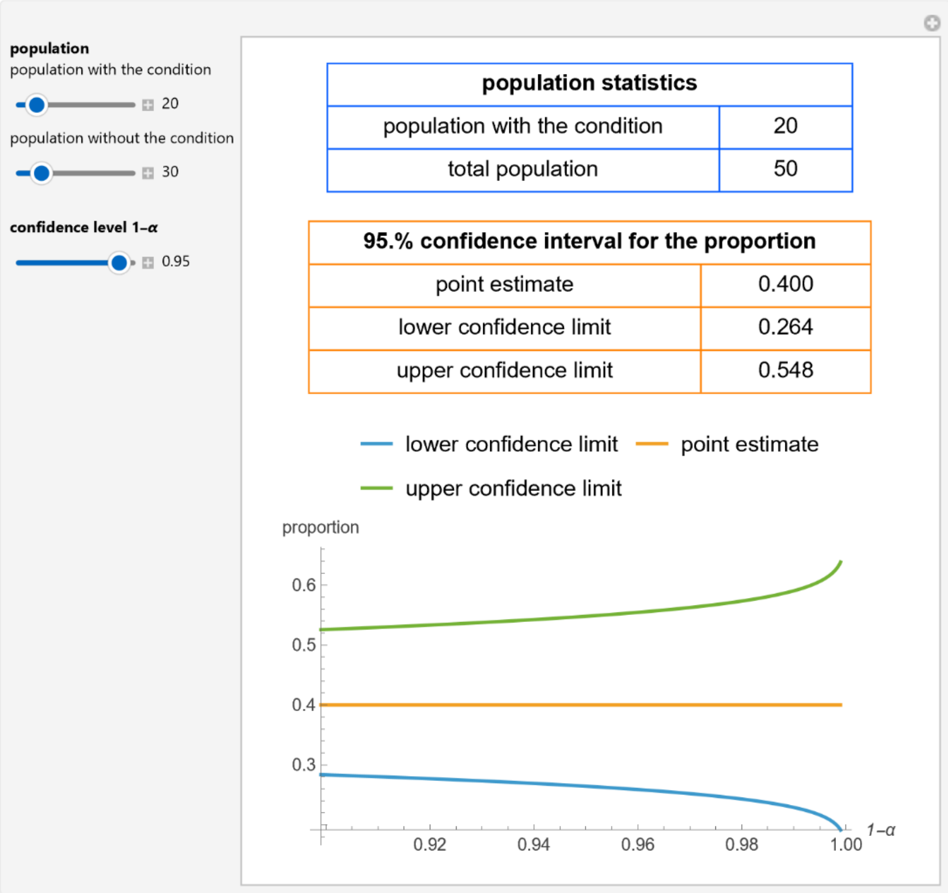


Figure 2: *Population statistics, point estimate, and 99% confidence interval for a single proportion of a population obeying a condition*, *as well as plots of the confidence limits versus confidence level*, *with the settings shown at the left.*



population statistics	
population with the condition	20
total population	50

95.% confidence interval for the proportion	
point estimate	0.400
lower confidence limit	0.264
upper confidence limit	0.548

Figure 3: *Population statistics, point estimate, and 95% confidence interval for a single proportion of a population obeying a condition, as well as plots of the confidence limits versus confidence level, with the settings shown at the left.*

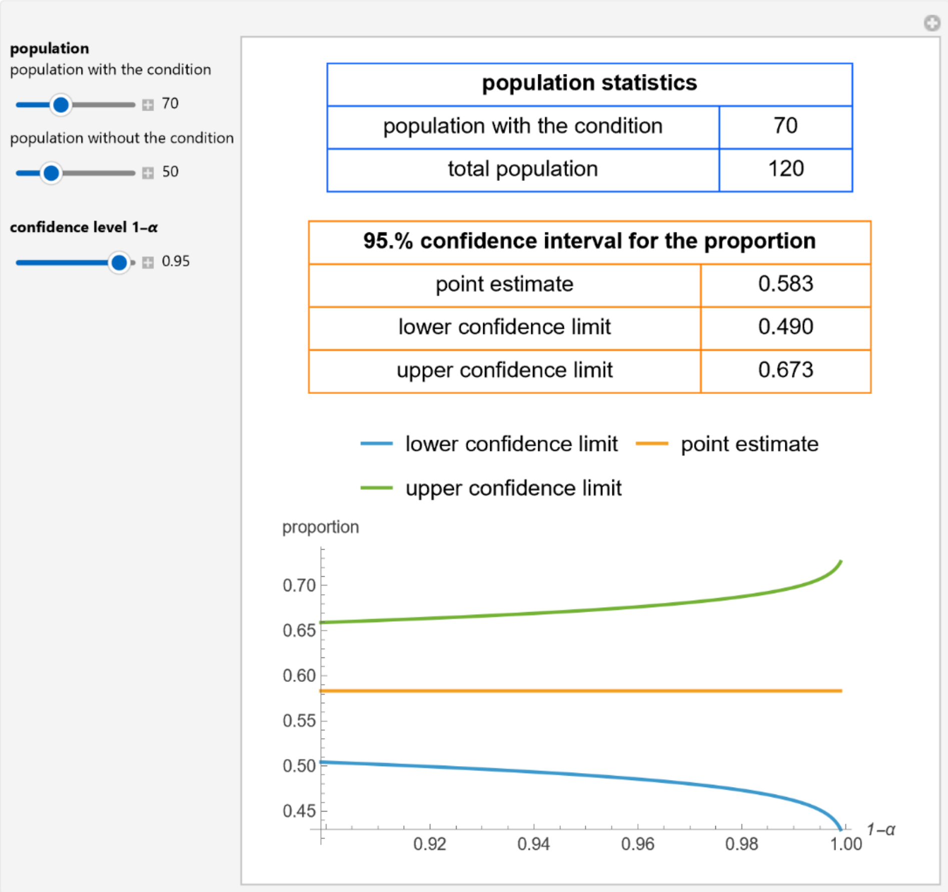


Figure 4: *Population statistics, point estimate, and 99% confidence interval for a single proportion of a population obeying a condition, as well as plots of the confidence limits versus confidence level, with the settings shown at the left.*

Details

Confidence intervals for individual proportions at a confidence level 1-α are calculated using the exact method based on the F-distribution [1].

Reference

[1] J. L. Fleiss, B. Levin and M. C. Paik. Statistical Methods for Rates and Proportions, 3rd ed., Hoboken, NJ: J. Wiley, 2003.

Demonstration

DOI

10.5281/zenodo.20381322

Source

Programming language: Wolfram Language
Availability: The updated source notebook is available at:
https://www.hcsl.com/Tools/Demonstrations/ExactConfidenceIntervalsForASingleProportion.nb

First published in 2018 as part of the Wolfram Demonstrations Project:
https://demonstrations.wolfram.com/ExactConfidenceIntervalsForASingleProportion/

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS
Other software requirements: Wolfram Player®, freely available at: https://www.wolfram.com/player/ or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.
System memory (RAM): 4 GB+ recommended.

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